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Role: AI/ML Intern

Module 3 — Cloth Segmentation AI

Overview:

Today I am working on Module 3 which is Cloth Segmentation AI. The main goal of this module is to separate different parts of an image like the person, clothing and background from each other using AI segmentation models.

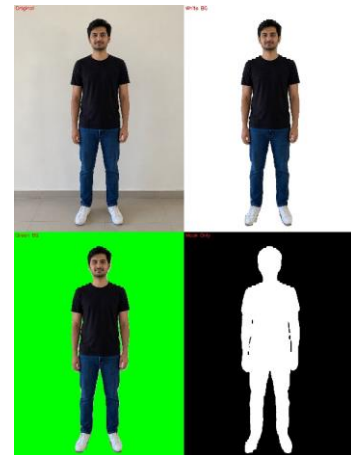
I started by researching different segmentation models and understanding how each one works. After that I tested and compared three basic segmentation models — MediaPipe, GrabCut and Watershed — on the same image to understand the difference in accuracy and output between traditional methods and AI based methods.

Model Used

1. **MediaPipe Selfie Segmentation** - a real time segmentation model made by Google that separates the person from the background using a deep learning model trained on selfie and portrait images.

Test 1 - Standing Straight

- Person standing upright facing the camera
- Result: Good accuracy
- The model correctly separated the person from the background



Test 2 - Sitting Pose

- Person sitting on a chair
- Result: Poor accuracy
- The model struggled to separate the chair legs from the background
- Black holes appeared in the foot and shoe area
- Chair and furniture were partially included in the person mask



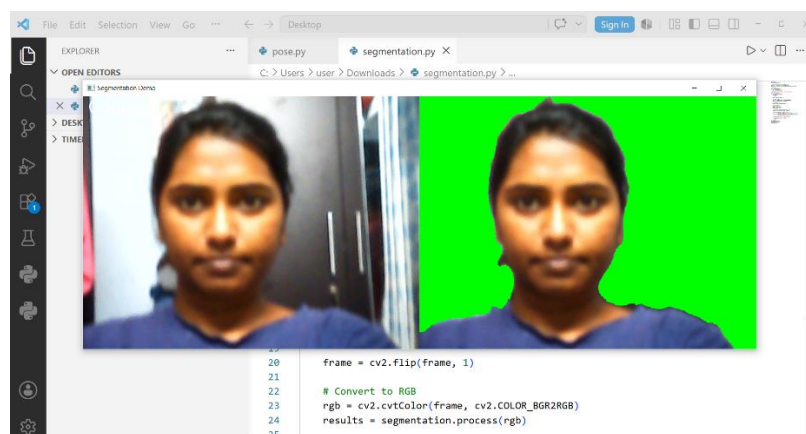
Test 3 – Selfie

I tested MediaPipe Selfie Segmentation on a live webcam feed. The code captures real-time video, generates a segmentation mask for each frame and replaces the background with green color.

Code:

```
measurment.py  angle_pose_detection.py  segmentation.py  segmentation_web.py X
segmentation_web.py > ...
1  import cv2
2  import mediapipe as mp
3  mp_selfie = mp.solutions.selfie_segmentation
4  segmentation = mp_selfie.SelfieSegmentation(model_selection=1)
5  cap = cv2.VideoCapture(0)
6  if not cap.isOpened():
7      cap = cv2.VideoCapture(1)
8  print("Segmentation Running... Press 'Q' to quit.")
9  while cap.isOpened():
10     ret, frame = cap.read()
11     if not ret:
12         break
13     frame = cv2.flip(frame, 1)
14     rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
15     results = segmentation.process(rgb)
16     mask = results.segmentation_mask > 0.5
17     bg = (0, 255, 0)
18     output = frame.copy()
19     output[~mask] = bg
20     combined = cv2.hconcat([frame, output])
21     cv2.putText(combined, "Original", (10, 30),
22                cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 255, 255), 2)
23     cv2.putText(combined, "Segmented", (frame.shape[1] + 10, 30),
24                cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)
25     cv2.imshow("Segmentation Demo", combined)
26
27     cv2.imshow("Segmentation Demo", combined)
28     if cv2.waitKey(1) & 0xFF == ord('q'):
29         break
30 cap.release()
31 cv2.destroyAllWindows()
32 segmentation.close()
33 print("Closed.")
```

Output:



Why Sitting Pose Failed

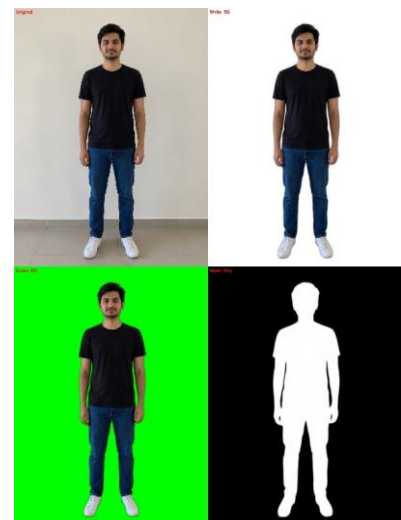
MediaPipe Selfie Segmentation is designed and trained mainly for selfies and upper body portraits. When the person is sitting with furniture like a chair in the frame the model gets confused between the person and the object. The chair legs and gaps between the person and chair are detected incorrectly causing holes and noise in the mask.

Model Used

2. **rembg (U2Net)** -an AI based background removal library built on deep learning that automatically detects the main subject and removes the background.

Test 1 — Standing Pose

- Person standing upright facing the camera
- Result: Excellent accuracy
- Background was completely and cleanly removed
- No holes or missing regions in the mask
- White and green background replacement looked natural



Test 2 — Sitting Pose

- Person sitting on a chair
- Result: Good background removal but chair included
- Background was cleanly removed
- The chair was included in the mask along with the person
- rembg cannot distinguish between the person and objects touching them
- Chair legs and seat were kept as foreground



Why Chair Was Included

rembg is a background removal model not a human specific model. It detects the main subject in the image as a whole. Since the person is sitting on the chair both are treated as one foreground object and kept together in the mask.

Deep Fashion Dataset

DeepFashion is a large fashion dataset with over 800,000 clothing images. It contains clothing photos, category labels, landmark points on clothes and paired images of the same clothing worn by a person and as a flat product photo. It is used to train fashion AI models to recognize and classify clothing items.

Dataset Used:

HR-VITON is a dataset specifically made for virtual try-on. Each sample has 4 images:

- Person photo
- Clothing flat lay image
- Cloth mask (white = clothing, black = background)
- Body segmentation map (different colors for different body parts)

How We Train Fashion AI Using Datasets

- Thousands of paired person and clothing images are collected
- Images are resized and prepared as inputs
- The model takes person image, clothing image and pose landmarks as input
- It predicts how the clothing looks on the person
- The prediction is compared with the actual output and the error is calculated
- The model keeps adjusting itself to reduce the error
- After many rounds of training the model learns to generate realistic try-on outputs

```
notebook56de6e89f5 Draft saved
File Edit View Run Settings Add-ons Help

+ [Icons] [Tools] Run All Code

=====
DATASET TESTING REPORT
=====

1. FOLDER STRUCTURE:
input/
  notebooks/
    mohitgupta12/
      hr-viton-try-on-clothes/
        HR-VITON/
          output/
            test/
              test/
                unpaired/
                  generator/
                    grid/
sync_batchnorm/
__pycache__/_
figures/
.git/
  info/
  refs/
    heads/
    remotes/
      origin/
hooks/
logs/
  refs/
    heads/
    remotes/
```



notebook56de6e89f5

Draft saved

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Run All

Code

Draft Session (6m)



```
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/_results__files: 1 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data: 1 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/image-densepose: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/agnostic-v3.2: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/openpose_img: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/image-parse-agnostic-v3.2: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/openpose_json: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/cloth: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/cloth-mask: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/image-parse-v3: 2032 files
/kaggle/input/notebooks/mohitgupta12/hr-viton-try-on-clothes/data/test/image: 2032 files
TOTAL FILES: 18405
```

3. CHECKING FOR CORRUPTED IMAGES:

Images Checked : 452

Corrupted Files: 0

Null/Bad Files : 0

4. IMAGE SIZE ANALYSIS:

Min Size : 716 x 1024

Max Size : 5400 x 15442

Avg Size : 906 x 1171

5. DATASET QUALITY:

Total images verified : 452

Corrupted : 0

Status: CLEAN – No corrupted files found

6. DISPLAYING SAMPLE IMAGES...

Sample Images from Dataset



Module 4 — Virtual Try-On AI

Notes:

Virtual Try-On is an AI system that digitally places clothing on a person's body without them physically wearing it. The user uploads a photo, and the AI automatically detects the body shape, warps the clothing to fit and generates a realistic output showing the person wearing that clothing.

Workflow / Pipeline

Person Photo + Clothing Image



Pose Estimation (MediaPipe)



Background Removal



Body Segmentation



Garment Warping



Image Synthesis



Final Try-On Output

Garment Warping

Garment warping is the process of stretching and adjusting the clothing image to match the body shape and pose of the person. It uses Thin Plate Spline (TPS) transformation which bends and deforms the clothing image so that:

- Collar aligns with neck
- Sleeves align with arms
- Hem aligns with waist

Without warping the clothing would look flat and unnatural on the body.

AI Fitting System

The AI fitting system estimates how well a garment fits a specific body. It uses body measurements extracted from pose landmarks and compares them with the garment dimensions to predict:

- Correct size
- Fit type -slim, regular or relaxed
- Whether the garment will look good on that body shape

Alignment Demo

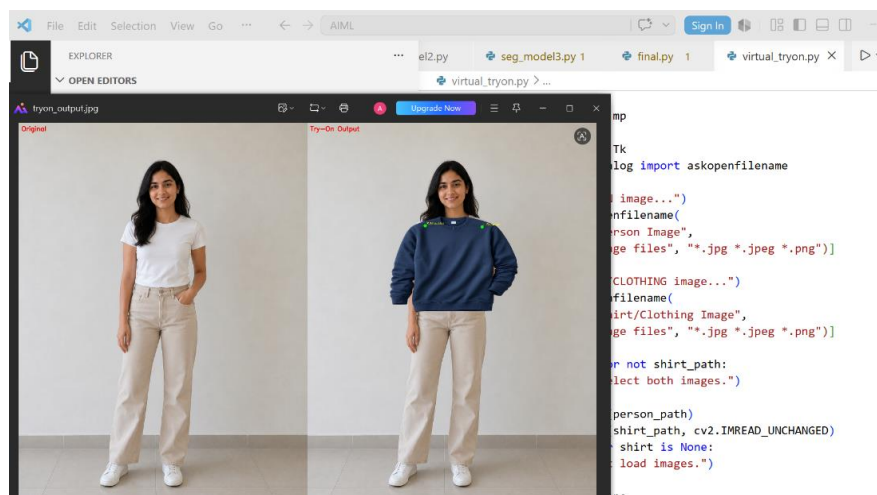
I built a simple virtual try-on demo that overlays a clothing image on a person photo using MediaPipe pose landmarks for alignment.

How It Works

- Person image and clothing image are selected using file picker
- MediaPipe detects body landmarks especially left and right shoulder positions
- Shirt width is calculated from shoulder to shoulder distance
- Shirt height is calculated from shoulder to hip distance
- Clothing image is resized and aligned to the shoulder landmarks
- Final output shows original and try-on result side by side

What I Observed

- The blue sweatshirt was successfully overlaid on the person
- Shoulder alignment was accurate - shirt aligned correctly to left and right shoulder landmarks
- The original white t-shirt is still visible underneath because we do not have cloth removal yet.



Pipeline:

First, pose detection identifies body points such as shoulders and arms. Then, background removal separates the person from the background. Finally, the shirt image is placed on the upper body using image overlay techniques to create the virtual try-on result.

